Factoring Special Forms: Squares

Answer Key

Algebra 1 Polynomials

Quick Answers

1. $(x-7)(x+7)$
2. $(x+5)^2$
3. $(3x-4)(3x+4)$
4. $(2x-3)^2$
5. $(x-4)(x+4)$
6. $(x+5)(2x-10)$
7. $(x+7)^2$
8. $(x-3)(x+3)$, x^2-6x+9
9. $(3x+4)(6x-8)$
10. $(x-3)(x+3)\left(x^2+9\right)$

Curriculum

Level: Algebra 1 Polynomials
HSA-SSE.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 12 tagged checks.

1. Factor x^2-49.
Step 1. Two terms with a minus sign between them: test the difference of squares. $x^2=(x)^2$ and $49=(7)^2$, so $A=x$ and $B=7$.
Step 2. $A^2-B^2=(A-B)(A+B)$ gives $(x-7)(x+7)$.
Step 3. Check by expanding: the two middle terms $+7x$ and $-7x$ cancel, leaving x^2-49 ✓ $\boxed{(x-7)(x+7)}$

2. Factor $x^2+10x+25$.
Step 1. Three terms, first and last both squares: $x^2=(x)^2$, $25=(5)^2$. Test the perfect-square pattern.
Step 2. The pattern needs a middle term of $2\cdot x\cdot 5=10x$, and that is exactly what is there.
Step 3. Because the middle sign is positive, the factor is a sum: $(x+5)^2$. $\boxed{(x+5)^2}$

3. Factor $9x^2-16$.
Step 1. $9x^2$ is a square as long as the coefficient is: $9x^2=(3x)^2$, and $16=(4)^2$, so $A=3x$ and $B=4$.
Step 2. $(3x)^2-(4)^2=(3x-4)(3x+4)$.
Step 3. Expanding gives $9x^2+12x-12x-16$, and the middle terms cancel ✓
$\boxed{(3x-4)(3x+4)}$

4. Factor $4x^2 - 12x + 9$.
Step 1. $4x^2 = (2x)^2$ and $9 = (3)^2$, so if this is a perfect square the roots are $A = 2x$ and $B = 3$.
Step 2. Test the middle term: $2AB = 2 \cdot 2x \cdot 3 = 12x$, and the trinomial has $-12x$. The sizes match and the sign is negative, so the factor is a difference.
Step 3. $4x^2 - 12x + 9 = (2x - 3)^2$. $\boxed{(2x-3)^2}$

5. Patio of side x with a 4 by 4 hot tub removed.
Step 1. The paved region is the big square minus the small square: $x^2 - 16$, which is $(x)^2 - (4)^2$.
Step 2. Difference of squares: $(x - 4)(x + 4)$.
Step 3. The factors are lengths in metres, and they are meaningful: cut the paved region into two rectangles and rearrange, and it measures $x - 4$ by $x + 4$. The area is unchanged, which is the geometric reason the identity is true. $\boxed{(x-4)(x+4)}$

6. Factor $2x^2 - 50$.
Step 1. $2x^2$ is not a perfect square, so the difference-of-squares pattern does not apply yet. Both terms share a factor of 2, so take it out first.
Step 2. $2x^2 - 50 = 2(x^2 - 25)$.
Step 3. Now the bracket is a difference of squares: $x^2 - 25 = (x-5)(x+5)$, so the complete factorization is $2(x - 5)(x + 5)$. $\boxed{2(x-5)(x+5)}$

7. Square mosaic of area $x^2 + 14x + 49$.
Step 1. A square's area is (side)^2, so factoring the area as a perfect square is what produces the side length.
Step 2. $x^2 = (x)^2$, $49 = (7)^2$, and the middle term must be $2 \cdot x \cdot 7 = 14x$ — it is.
Step 3. The area is $(x + 7)^2$, so the side of the mosaic measures $x + 7$ centimetres.
$\boxed{(x+7)^2}$

8. Error analysis: is $x^2 - 9 = (x - 3)^2$?
Step 1. Expand the student's answer: $(x - 3)^2 = x^2 - 3x - 3x + 9 = x^2 - 6x + 9$.
Step 2. That is not $x^2 - 9$: it has an extra $-6x$ and the constant has the wrong sign. So the student's factorization is wrong.
Step 3. $x^2 - 9$ has two terms, not three, so the pattern it fits is the difference of squares: $x^2 - 9 = (x - 3)(x + 3)$.
Manual review: the written sentence is an open response. Accept any wording saying the student used the perfect-square pattern on a two-term expression, or that squaring a binomial produces a middle term the original never had. $\boxed{x^2 - 6x + 9 \quad \text{and} \quad (x-3)(x+3)}$

9. Factor $18x^2 - 32$.
Step 1. Neither $18x^2$ nor 32 is a perfect square, but both are even: the greatest common factor is 2.
Step 2. $18x^2 - 32 = 2(9x^2 - 16)$, and now the bracket is a difference of squares with $A = 3x$, $B = 4$.
Step 3. $2(3x - 4)(3x + 4)$. Expanding back gives $2(9x^2 - 16) = 18x^2 - 32$ ✓
$\boxed{2(3x-4)(3x+4)}$

10. Factor $x^4 - 81$.

Step 1. $x^4 = (x^2)^2$ and $81 = (9)^2$, so this is a difference of squares with $A = x^2$ and $B = 9$.
Step 2. $x^4 - 81 = (x^2 - 9)(x^2 + 9)$.
Step 3. The first bracket is itself a difference of squares: $x^2 - 9 = (x - 3)(x + 3)$. The second is a *sum* of squares, which does not factor over the integers, so the factorization stops here. $\boxed{(x-3)(x+3)(x^2+9)}$